

Ground Truth

AP Physics C · E&M, derived

A lightning research station on a salt flat: a sixty-metre mast, a launch rail, and a storm coming in.

45	23	3.1	11/11	5	24 · 19 · 2
STOPS	CHOICE (51%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

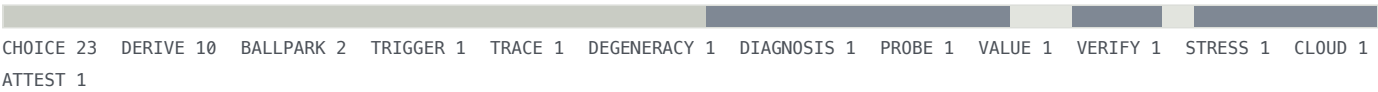
READING Where this campaign stands

The cleanest curriculum in the set outside Headwater: **all eleven equations computed**, no debt at all. It is also the second-worst format mix in the catalogue at 23 CHOICE out of 45 — which is exactly the pair of facts that killed the diversity gate. Nothing here is broken. The finding is six select-only mechanism concepts: fields of sheets and spheres, shielding, equipotentials, force on a current, the LR transient, and energy in the magnetic field — all currently reached by picking from four.

PROTECTED — DO NOT CONVERT

All ten DERIVE stops stay. Ground Truth is AP Physics C E&M taught in ten derivations, and that is the game. No slate below touches one.

NOW The mix as it ships



CURRICULUM What the syllabus is not getting

No equation debt. Every equation the syllabus lists is computed by a question somewhere in the campaign.

TIER Concepts the player only ever picks from a list

#	MECHANISM	CONCEPT	STOPS
5	Field inside a conductor, and shielding		6, 24, 26, 39
9	Equipotentials, and what they mean on the ground		8, 14
22	Force on a current in a field, and forces between conductors		24
28	The LR transient, and inductive volts against resistive volts		30
29	Energy in the magnetic field, and where a strike puts it		24

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects. The syllabus also lists 1 concept no stop touches at all — **Biot–Savart, and when the integral is worth doing** — which is expected, and is a different conversation from this one.

LOB STRONG

SHOT · Launch Control

This is the one game where LOB is not a compromise. The site has a launch rail and the science is rocket-triggered lightning: you fire a wire-trailing rocket into a charged cloud base. Angle and charge against a mark is the actual operation, and the withheld quantity is honest — nobody knows the payout well enough to compute the strike, which is **why** the technique is empirical. Everywhere else in the catalogue LOB teaches "it cannot be computed" as a lie. Here it is the finding.

SERVES triggered-lightning technique, and the empirical half of a derived course

NEEDS marks ordered outward by distance, and a rigging quantity that is genuinely unknowable

SWEEP STRONG

stop 8 or 14 · COUPLE / EARTH

Walk a probe out across the earthing compound and plot potential against distance: equipotentials are a shape, and concept 9 is select-only. "One foot and the other" is a CHOICE about step potential, which is the same curve read at two points.

SERVES concepts 9 and 25 — equipotentials, step-and-touch

NEEDS the response plotted only where the player looks

HOLD STRONG

BANK · Impulse Hall

Charge a capacitor bank to a target and hold it while leakage drains it and the storm window closes. The LR transient — concept 28, select-only — is a rate you feel rather than a formula you recognise.

SERVES concept 28, inductive volts against resistive volts

NEEDS a do-nothing run must fail

SPOT STRONG

a MAST or FIELD stop

A storm-safety instruction replaced mid-run: sixty metres of instrumented mast on a salt flat with a cell coming in is the most legitimate reason in the set for an instruction to change without ceremony.

SERVES operating discipline under a moving hazard

TRIAL STRONG

the salt flat

The flattest site in the catalogue, with a mast, an earthing compound, an outstation and a launch rail — and a storm arriving. Driven order: earth before you touch, screen before you connect.

SERVES concept 5, shielding, and concept 25

RISK world work

TALLY STRONG

a SEIS-equivalent SHOT stop

Strikes accumulate and the statistic firms up. A storm season is exactly a case where the early number moves for the wrong reason.

SERVES measurement statistics on a rare event

BELT REJECT

Nothing here is sorted. Rejected on subject.

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: `curriculumDelivery.mjs groundtruth --snapshot before.json`, then `--against before.json`.

Minimum honest 3 STOPS · ONE SITTING

6	d2	MAST	CHOICE → PROBE	Inside the sheet inside the sheet — read station by station instead of picked
8	d3	COUPLE	CHOICE → SWEEP	One foot and the other step potential as a curve across the compound
24	d8	COUPLE	CHOICE → CHAIN	The cabinet is a loop the cabinet as a loop, and the link that governs the current

CHOICE **23 → 20** (44%) · moves **24·19·2 → 21·20·4** · effective formats **3.1 → 3.9** · **2** new operate stops

Balanced 6 STOPS · THE RECOMMENDATION

6	d2	MAST	CHOICE → PROBE	Inside the sheet
8	d3	COUPLE	CHOICE → SWEEP	One foot and the other
24	d8	COUPLE	CHOICE → CHAIN	The cabinet is a loop
30	d10	BANK	CHOICE → HOLD	Making a microsecond on purpose making a microsecond on purpose, held against leakage
39	d13	MAST	CHOICE → SPOT	What to put on a microsecond what to put on a microsecond, under an instruction that changes
2	d1	FIELD	CHOICE → BALLPARK	A spinning shutter and a sign the spinning shutter, computed

CHOICE **23 → 17** (38%) · moves **24·19·2 → 18·21·6** · effective formats **3.1 → 4.9** · **4** new operate stops, 2 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

6	d2	MAST	CHOICE → PROBE	Inside the sheet
8	d3	COUPLE	CHOICE → SWEEP	One foot and the other
24	d8	COUPLE	CHOICE → CHAIN	The cabinet is a loop
30	d10	BANK	CHOICE → HOLD	Making a microsecond on purpose
39	d13	MAST	CHOICE → SPOT	What to put on a microsecond
2	d1	FIELD	CHOICE → BALLPARK	A spinning shutter and a sign
12	d4	SHOT	CHOICE → LOB	The afternoon Vero is right about the launch rail — the one honest LOB in the catalogue
14	d5	EARTH	CHOICE → TALLY	Two conductors, two hundred metres two conductors, two hundred metres, and enough strikes to say so
3	d1	EARTH	CHOICE → TRIAL	Twenty-five ohms, at what the flat, driven — earthing before touching

CHOICE **23 → 14** (31%) · moves **24·19·2 → 15·21·9** · effective formats **3.1 → 6.2** · **7** new operate stops, 4 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent.
2. Snapshot. 3. Edit the book, never the generated content — tools/import-book.mjs books/groundtruth.yml groundtruth --verify. 4. npm run traps: every instrument's trap must fire. 5. npm run check groundtruth and npm run drive groundtruth, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. 6. --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from `themes/groundtruth/content/` via `engine/dev/curriculumDelivery.mjs`; concept and equation lists authored in `tools/syllabus.js`. Format verdicts and slates are editorial. Run it with `THEME=groundtruth npm run dev`.